

Introduction to structure-based drug design

A practical guide

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SCHRÖDINGER.

Drug discovery to drug development pipeline



- Preexisting target
- Literature target
- Genomics
- Proteomics

Identification

- HTS
- De novo design
- Virtual screening

Optimization

- SAR studies
- In vivo efficacy
- ADMET studies

- Bioavailability studies – ADME
- Long-term toxicity
- In vivo efficacy
- Synthesis scale-up
- Efficacy
- Bioavailability
- Dose
- Safety assessments
- Drug interactions

Time-consuming

- 2-5 yrs discovery; ~13.2 yrs development

Cost-intensive

- \$1.1bn (1995-2000) - \$1.7bn (2002-2002)¹
- Total R&D costs for Discovery²
 - 25% (UK); 33% (US)

1. Glibert . J. et al. (2003) In Vivo, the Business & Medicine Report **21** (10)

2. Data from CMR Compendium, (2002) PhRMA Survey (2002)

Virtual screening

- Virtual screening is a fast and cost effective method to generate leads for drug design
- Alternative or complementary to high throughput screening
- Virtual screening requires careful planning, execution, and processing

Ensemble docking

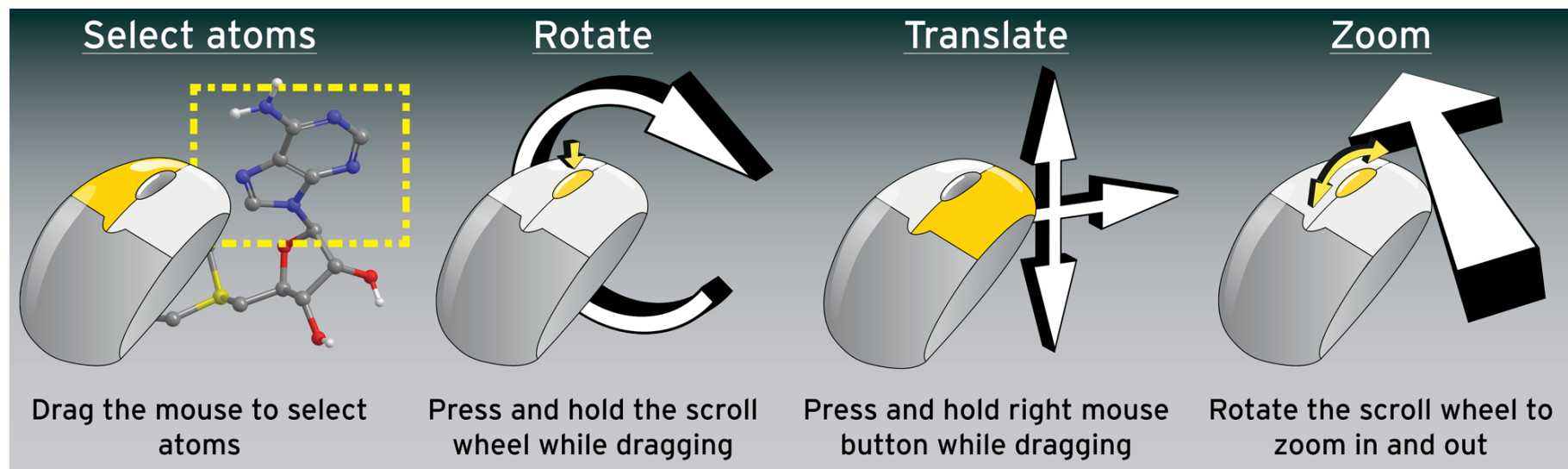
- Can leverage crystal structures, when available
 - No need to explicitly choose flexible residues of interest
 - Can account for multiple movements in one structure
- MD can be used in addition to or in place of crystal structures
 - Allows for more flexibility

Today's goals

- Become oriented with Maestro user interface and some popular tools
- Set up and run a self-docking job with Glide to validate our target model
 - dock a known binder to our target structure
- Learn how to use docking analysis tools
- Empower you to explore additional tools for virtual screening, addressing receptor flexibility, and other tools that will help to address your research questions

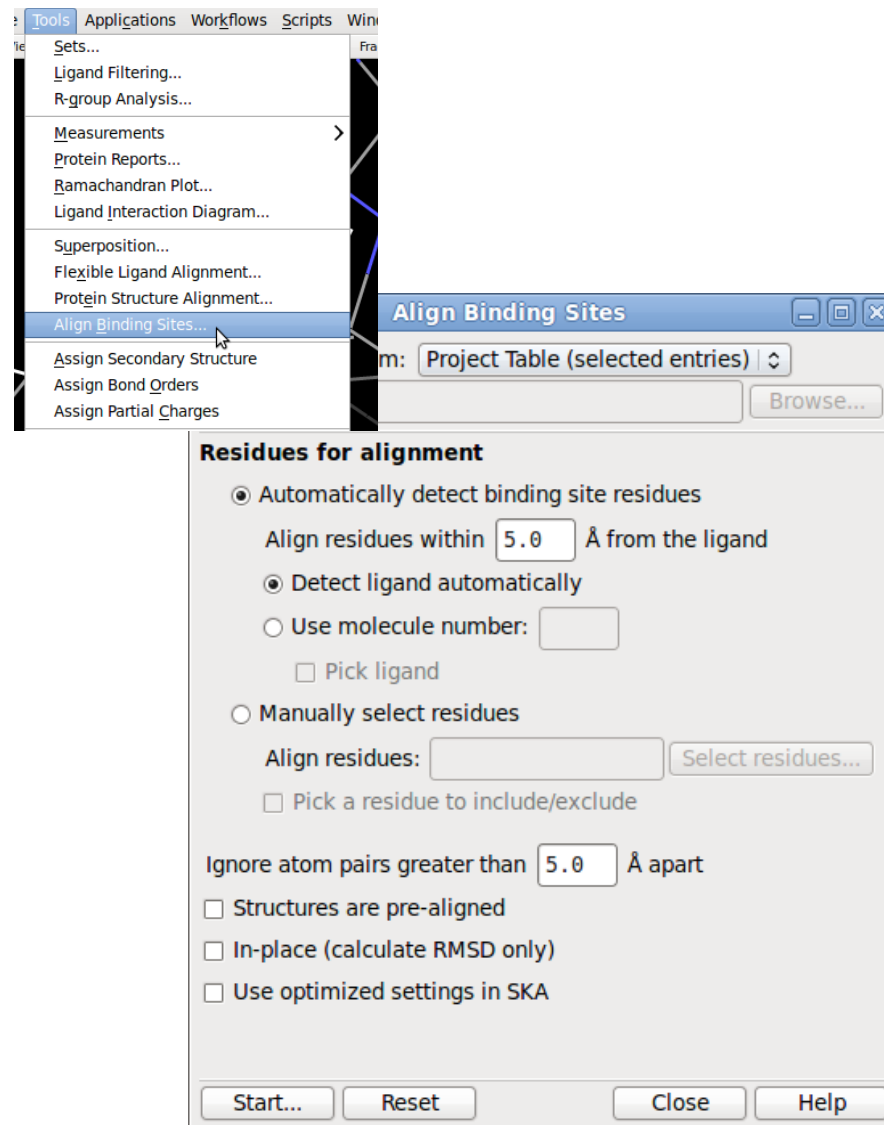
Let's get oriented with Maestro, the user interface

- Setting up the Workspace, toggling on your toolbars
- Saving the project
- Using the 2D sketcher
- Using the Project Table
- Learning the mouse controls

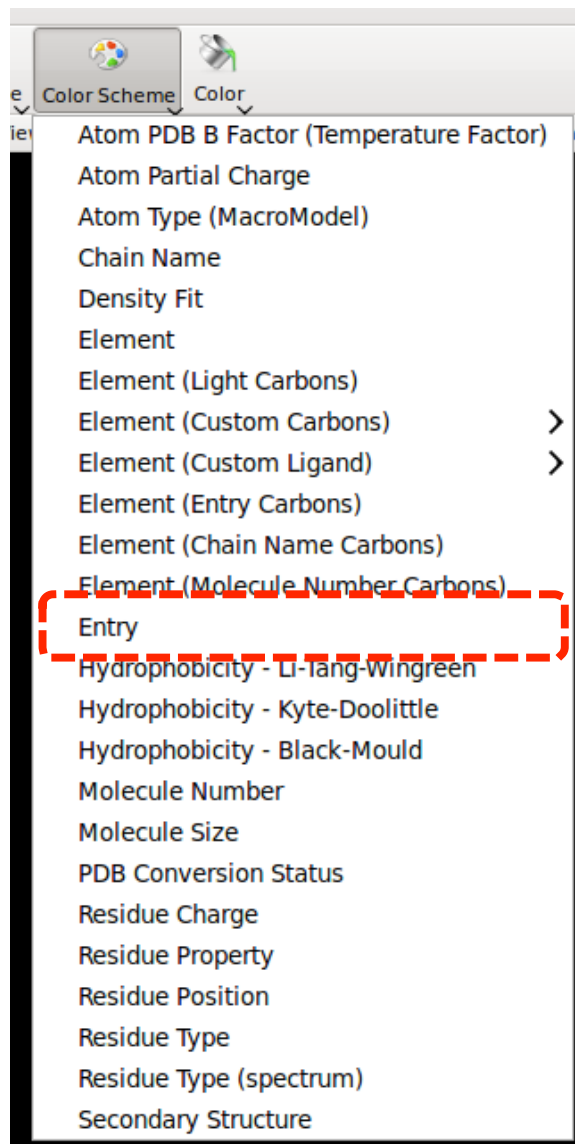


Binding site alignment

- Use Get PDB to bring 1LCZ and 1STP into the Workspace
- Choose Tools > Align binding sites...
- Select two or more Project Table entries to align. Structures will be aligned against the topmost entry in the Project Table.



Compare active sites from different PDB structures



- Compare active sites from different PDB structures to reveal induced fit effects or look for water molecules that should be retained during protein preparation:
 - Align binding sites
 - Color by entry
 - Apply Workspace Style (or display only the ligand and residues near the ligand)
 - Create and clip binding site surfaces to observe pockets that are present in one chemotype but not another.



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Other handy orienting tools

- Window > Sequence viewer
- Hovering over an atom in the Workspace will reveal information about that atom at the bottom of the Workspace.
- Use Ctrl-F to find a specific residue

Pre-screening considerations

- Consult the literature

RCSB PDB PROTEIN DATA BANK **PDB-101** A MEMBER OF THE PDB

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STRUCTURAL ORIGINS OF HIGH-AFFINITY BIOTIN BINDING TO STREPTAVIDIN

DOI:10.2210/pdb1stp/pdb

Primary Citation

Structural origins of high-affinity biotin binding to streptavidin.
[Weber, P.C.](#), [Ohlendorf, D.H.](#), [Wendoloski, J.J.](#), [Salemme, F.R.](#)

Journal: (1989) Science **243**: 85-88

PubMed: [2911722](#) [Open](#)

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PubMed Abstract:
The high affinity of the noncovalent interaction between biotin and streptavidin forms the basis for many diagnostic assays that require the formation of an irreversible and specific linkage between biological macromolecules. Comparison of the refined crystal structures of apo and...
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Molecular Description Hide

Classification: Biotin Binding Protein [Open](#)

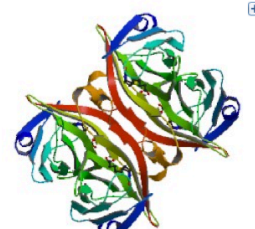
Structure Weight: 16749.31 [Open](#)

Molecule: STREPTAVIDIN COMPLEX WITH BIOTIN

1STP

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Biological Assembly [?](#)



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Biological assembly 1 assigned by authors and generated by PISA,PQS (software)

Downloadable viewers:

Pre-screening considerations

- Before spending time on a large virtual screen, attempt to answer the following questions to validate your model:
 - (1) Can I reproduce the experimentally determined binding modes for known ligands? (We'll do this for 1STP today.)
 - (2) Can Glide retrieve known active compounds from a decoy set?

Can I reproduce the experimentally determined binding modes for known ligands?

- (1) Extract the bound ligand from the experimentally derived structure you have prepared
 - (2) Re-dock the bound ligand using Glide
- If there are unexpected results:
 - Do different treatments of active site water molecules result in significantly different GlideScores?
 - Should I apply docking constraints in order to force Glide to return crucial receptor-ligand interactions?

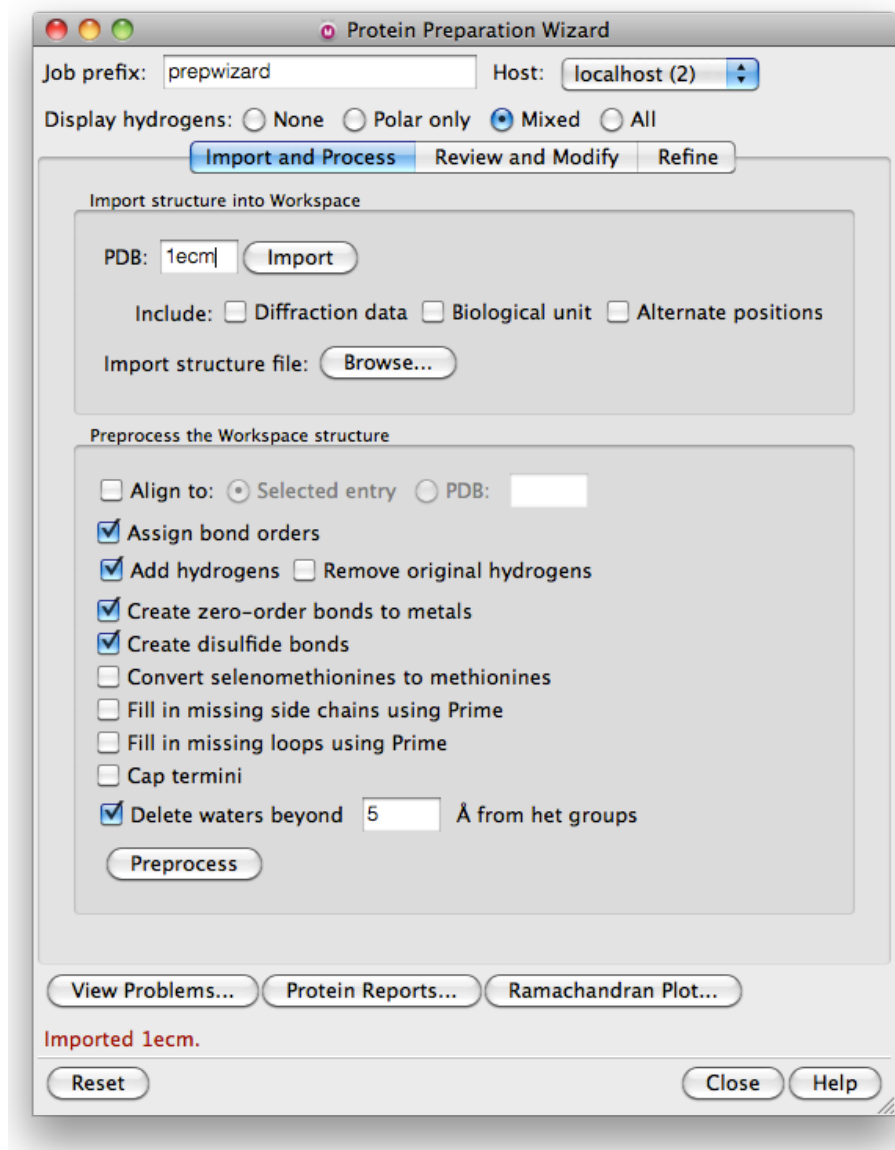
Can Glide retrieve known active compounds from a decoy set?

- (1) Download the 1000-compound decoy library from the Glide web page
 - (2) Seed the decoy library with known active compounds
 - (3) Dock the seeded decoy library against your target structure
- Are active compounds ranked near the top of the database?

Preparing for docking

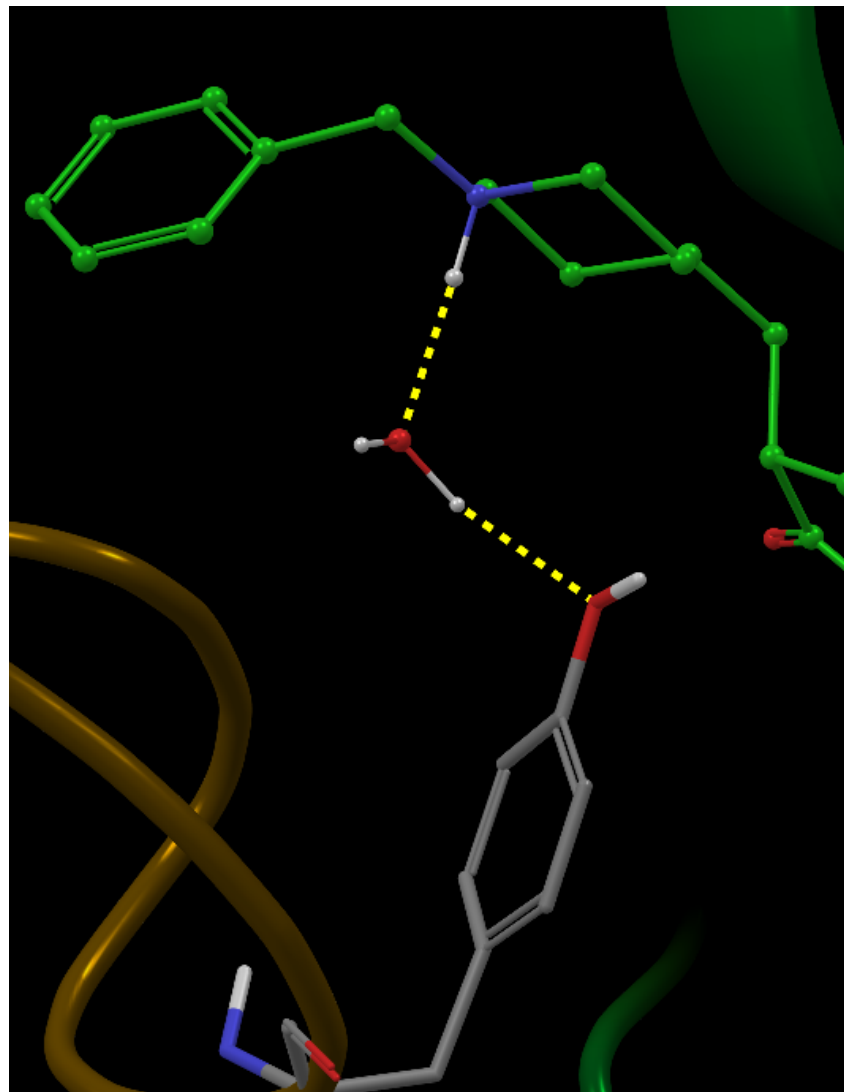
- Things you'll need:
 - Prepared target structure > Protein Preparation Wizard
 - Prepared ligand(s) > LigPrep

Preparation of target structure (Protein preparation wizard)



- Preparing the PDB structure
- Assigning protonation states and tautomers for the ligand
- Hydrogen bonding network optimization, Asn, Gln flips
- Restrained minimization

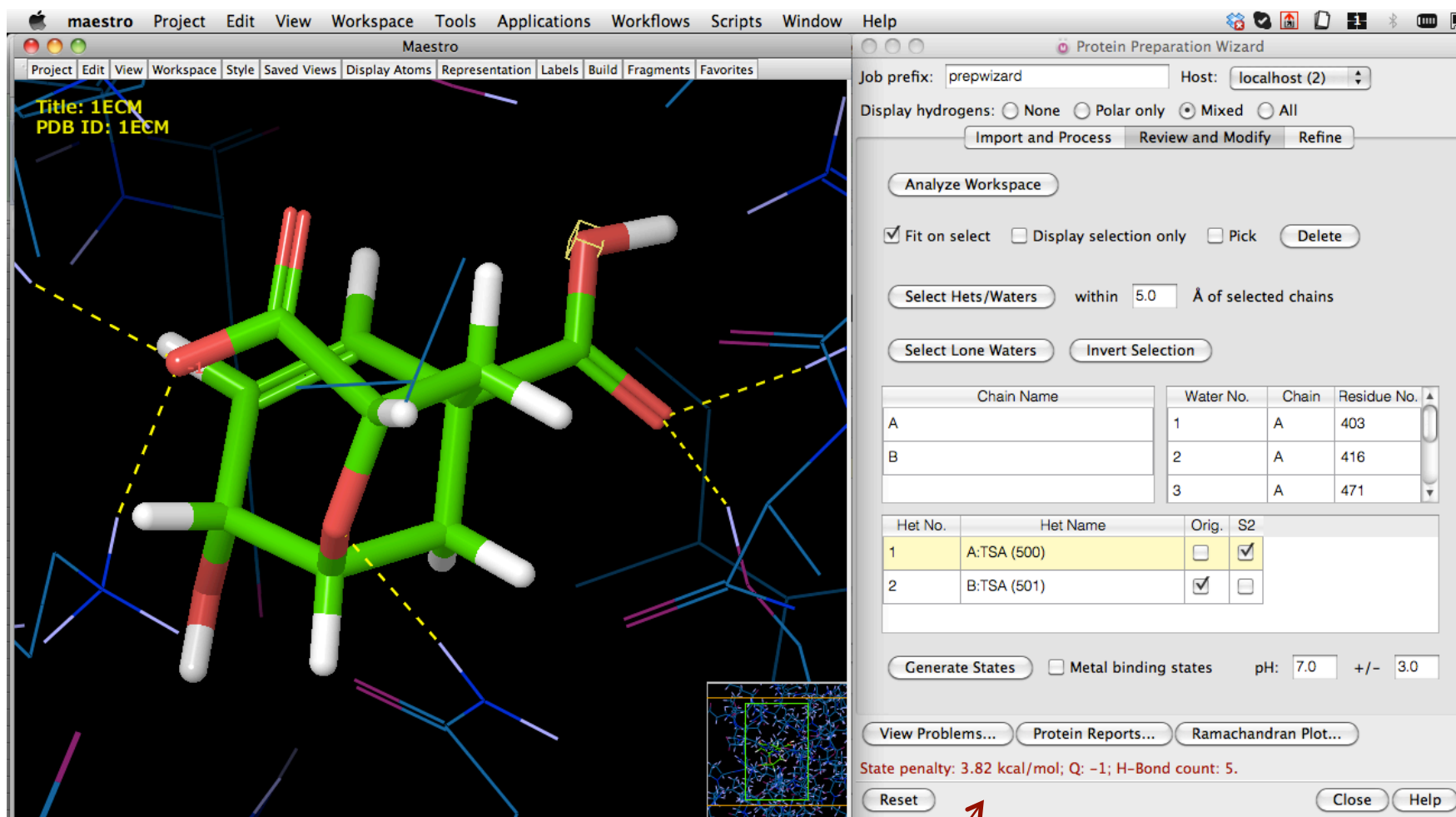
Dealing with water molecules in the active site



- As a general rule of thumb, consider retaining “structural waters” that mediate receptor-ligand binding
- Structural water: water molecule that forms an H-bond to the receptor, and another H-bond to the ligand
- However, even these water molecules may be removed if ligands are capable of displacing the water molecules

Ligand States with Epik

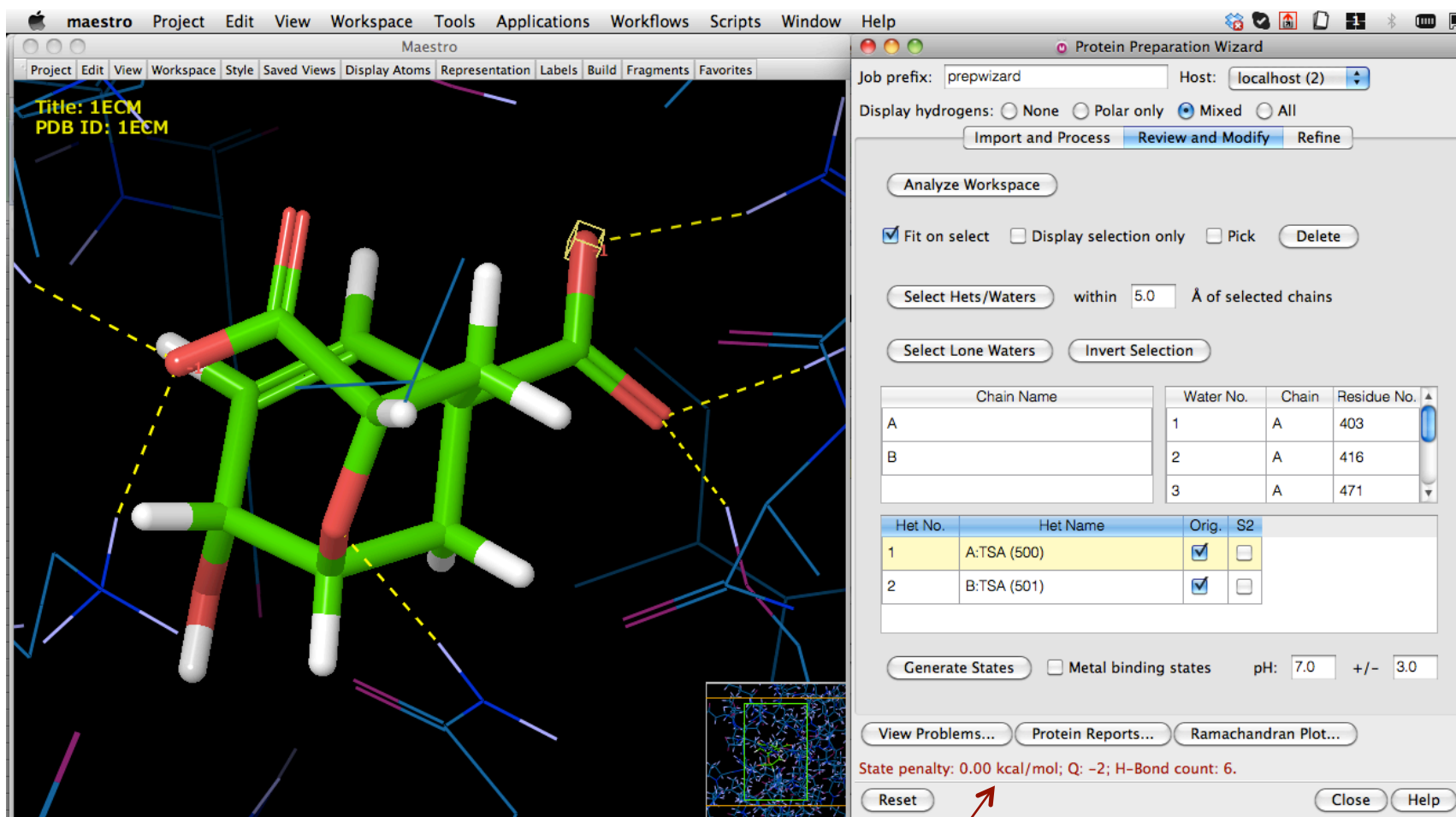
- Available states within pH range with energetic penalties



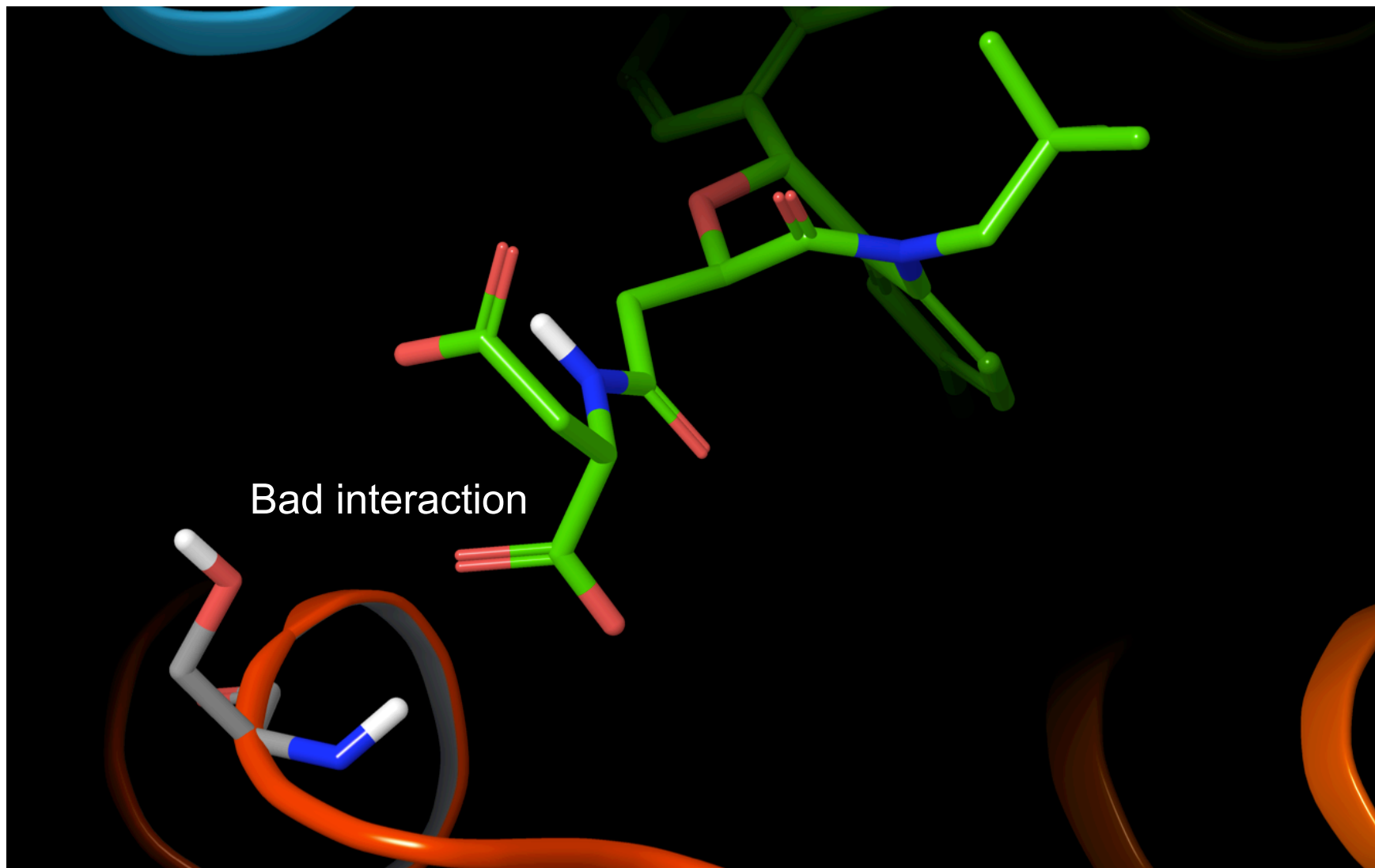
3.82 kcal/mol **SCHRÖDINGER.**

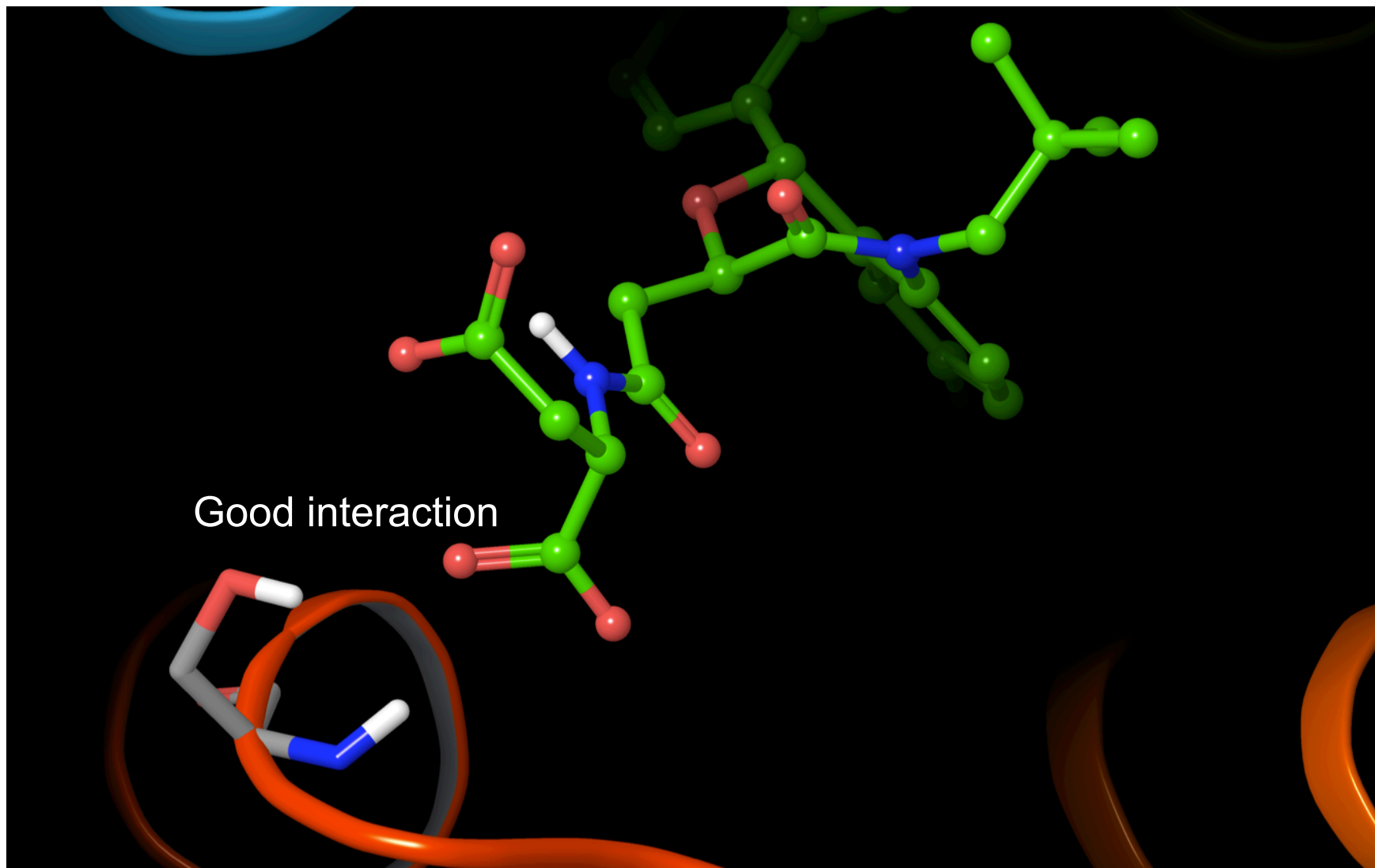
Ligand States with Epik

- Available states within pH range with energetic penalties



0.00 kcal/mol **SCHRÖDINGER.**





Protein Minimization

- Differences are subtle
- Small atomic deviations of ~ 0.1 Å can have an impact on enrichments
- Allowing for a known active to fit well tends to allow for other known actives to fit too
 - If packing around a core for a co-crystallized ligand is suboptimal then it is likely to be suboptimal for similar compounds too

Preparing for any docking calculation, including self-docking

- Things you'll need:
 - Prepared target structure > Protein Preparation Wizard
 - Prepared ligand(s) > LigPrep

Goals of ligand library preparation with LigPrep

- Convert 2D compounds into 3D compounds
- When beginning with 2D structures, honor the specified chirality (when present)
- Enumerate ligand ionization states and tautomers
- Remove unnecessary counterions (“Desalt”)
 - For example, if an input compound is shown as a hydrochloride salt, LigPrep will remove the chloride ion from the structure.
- Calculate the state penalties (i.e., relative energies) of different ionization states and tautomers for individual ligands
 - State penalties are calculated within LigPrep using a program called Epik
 - Glide will account for these state penalties in the DockingScore
 - Accounting for state penalties reduces false positives – this significantly improves Glide enrichment

Running LigPrep

- Generate states with Epik
- Stereoisomers:
 - Use default setting for 2D input
 - Use “Determine from 3D structure” for 3D input
- If docking against a metalloprotein, use the option “Add metal binding states”

The screenshot shows the LigPrep window with the following settings:

- Use structures from: File
- File name: itory/glide_docking/2d-ligands.sdf
- Filter criteria file: (empty)
- Force field: OPLS_2005
- Ionization:
 - ☐ Do not change
 - ☐ Neutralize
 - ☒ Generate possible states at target pH: 7.0 +/- 2.0
- Using: ☐ Ionizer ☒ Epik
- ☐ Add metal binding states
- ☐ Include original state
- ☒ Desalt ☒ Generate tautomers
- Stereoisomers**
 - Computation:
 - ☒ Retain specified chiralities (vary other chiral centers)
 - ☐ Determine chiralities from 3D structure
 - ☐ Generate all combinations
 - Generate at most: 32 per ligand
 - Generate low energy ring conformations: 1 per ligand
 - Output format: ☒ Maestro ☐ SDF

Buttons at the bottom: Start..., Close, Help

Setting up a Glide docking calculation

What information does Glide need before docking?

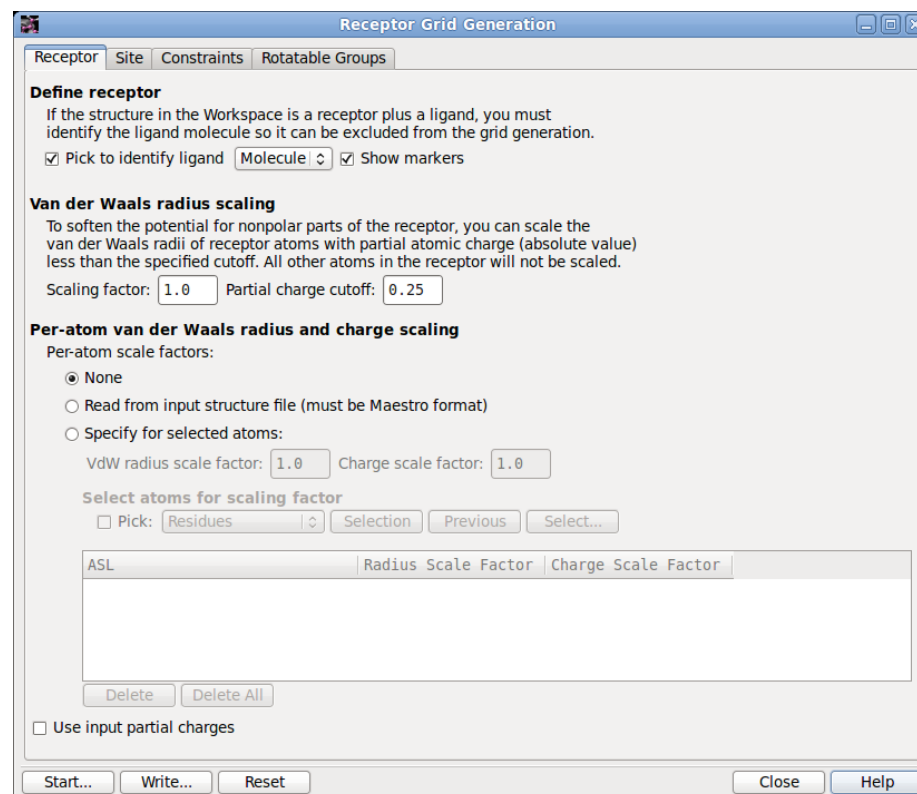
- Glide needs to know:
 - What target structure will I be docking against (i.e., what grid files will you be using)?
 - What ligands do I want to dock?
 - What docking mode should I use?
- Goal:
 - Use Glide to correctly position your ligands within the target's active site
 - Evaluate the energetics of protein-ligand binding
 - Examine a “Poseviewer” file that contains docked ligands ranked according to their DockingScore

Using “grid” files with Glide

- A Glide docking experiment is a two-step procedure:
 - (1) Generate grids
 - (2) Dock ligands against the grids that you created for your target
- The grid files are a grid-based abstraction of the specified binding site
- You only need to generate grids once for any 3D target structure – the grid files can be used repeatedly for multiple docking runs
- However, you may perhaps wish to generate multiple grids for your target structure using:
 - Different grid generation settings
 - Different protein preparation strategies (e.g., retain different sets of active site waters, or different plausible H-bonding networks)

Step 1: Grid generation

- In the main Maestro menu, choose Applications > Glide > Receptor Grid Generation
- Setting up a grid generation is no more complicated than clicking on the ligand in your active site, and clicking “Start” to submit the job
- Advanced options are in the next three tabs of the Grid Generation interface



Step 2: Dock ligands with specified mode

Ligand Docking

Settings | Ligands | Core | Constraints | Torsional Constraints | Similarity | Output

Receptor grid
Specify the receptor grid you want to use for docking.
Receptor grid base name:
☐ Display receptor ☐ Show grid boxes

Docking
Precision:
☐ HTVS (high throughput virtual screening)
☐ SP (standard precision)
☒ XP (extra precision)
Options:
☒ Dock flexibly
☒ Sample nitrogen inversions
☒ Sample ring conformations
☐ Include input ring conformation
Amide bonds:
☐ Dock rigidly
☐ Refine (do not dock)
☐ Score in place (do not dock)
☒ Write XP descriptor information
☒ Add Epik state penalties to docking
☐ Use enhanced planarity force field parameters

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☒ Write XP descriptor information
☒ Add Epik state penalties to docking score
☐ Use enhanced planarity force field parameters

- Choose your grid files and a scoring mode.
 - Best practice: Always make sure to calculate the Docking Score by adding Epik state penalties.
 - Best practice: Always calculate XP scoring terms when running an XP docking job.

Choose your ligands to dock

- Choose to dock a saved ligand library, selected entries in Maestro's Project Table, or a single ligand in the Maestro workspace.

The screenshot shows the 'Ligands' tab in the Maestro software interface. At the top, there is a row of tabs: 'Settings', 'Ligands' (which is selected), 'Core', 'Constraints', 'Torsional Constraints', 'Similarity', and 'Output'. Below the tabs, the section is titled 'Ligands to be docked'. A text block states: 'We strongly recommend that you prepare the ligands before docking (for example, with LigPrep or MacroModel)'. Below this, it says 'Use ligands from:'. There are three radio button options: 'File', 'Selected entries' (which is selected), and 'Workspace'. Under the 'File' option, there is a 'File name:' text box followed by a 'Browse...' button. Below the 'File name' box, there is a 'Range:' section with input boxes for '1' and '1000', a 'to:' label, and a checked checkbox followed by the word 'End'. The 'Selected entries' option is highlighted with a blue bar.

Viewing Glide docking results from our self-docking job

Browsing a poseviewer file in the Project Table

- Poseviewer mode:

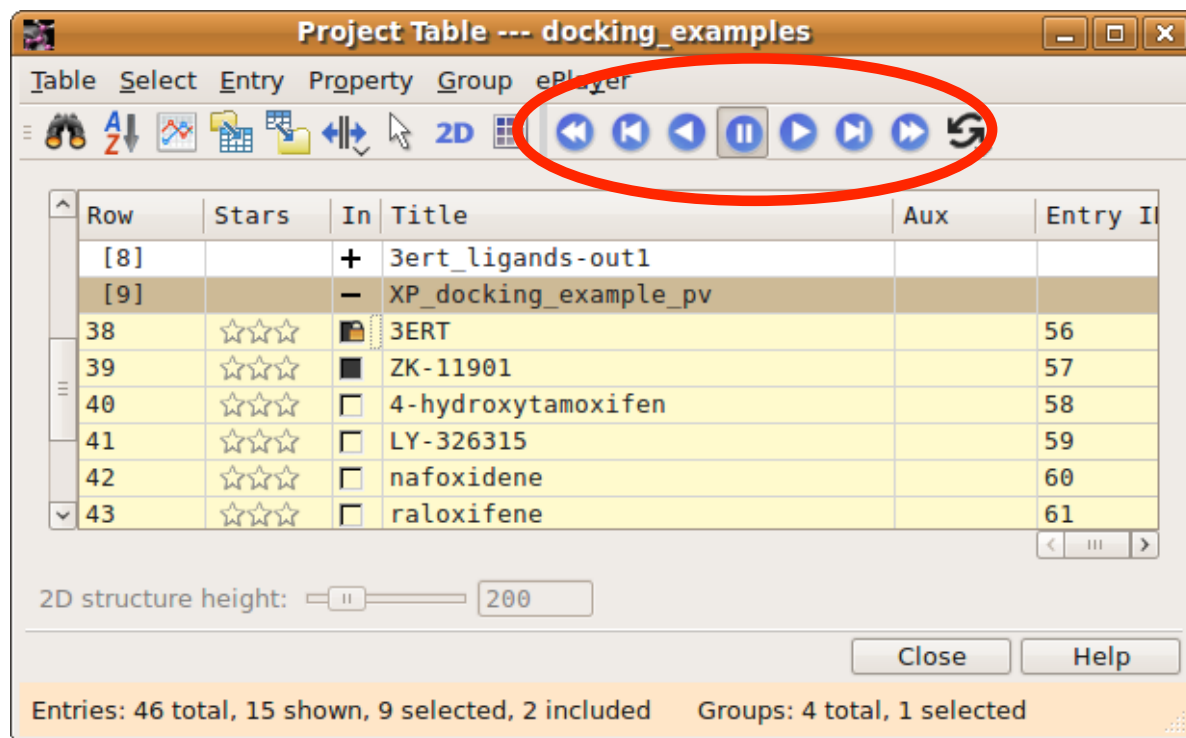
- Is automatically enabled whenever you import a poseviewer file
- “Fixes” or “locks” the receptor structure in the Workspace, allowing you to browse docked ligands
- Can be turned on by right-clicking the name of an entry group and choosing “View Poses > Setup”

[33]	—	[2] - 3ert_xp_docking_pv
5		3ERT
6		4-hydroxytamoxifen
7	☐	ZK-11901
8	☐	ICI-164384
9	☐	ICI-164384
10	☐	LY-326315
11	☐	nafoxidene
12	☐	ZK-11901
13	☐	ICI-164384

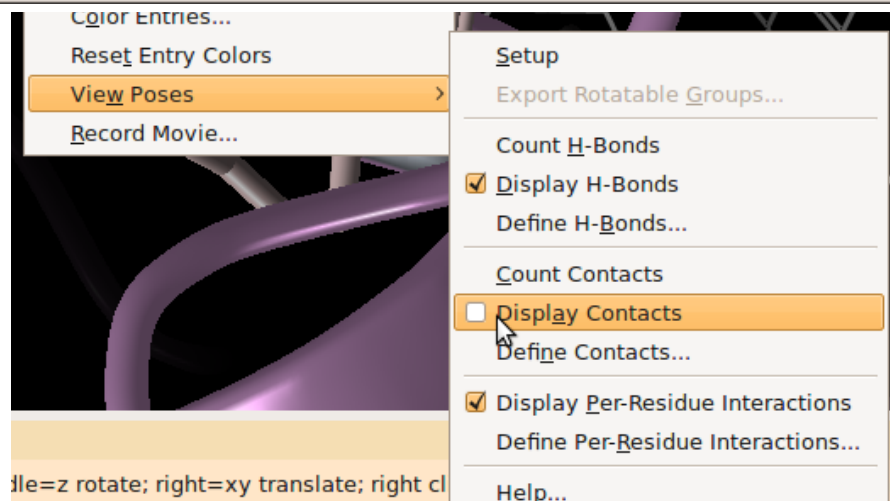
Viewing Glide results

- Receptor structure is fixed in the Workspace by default

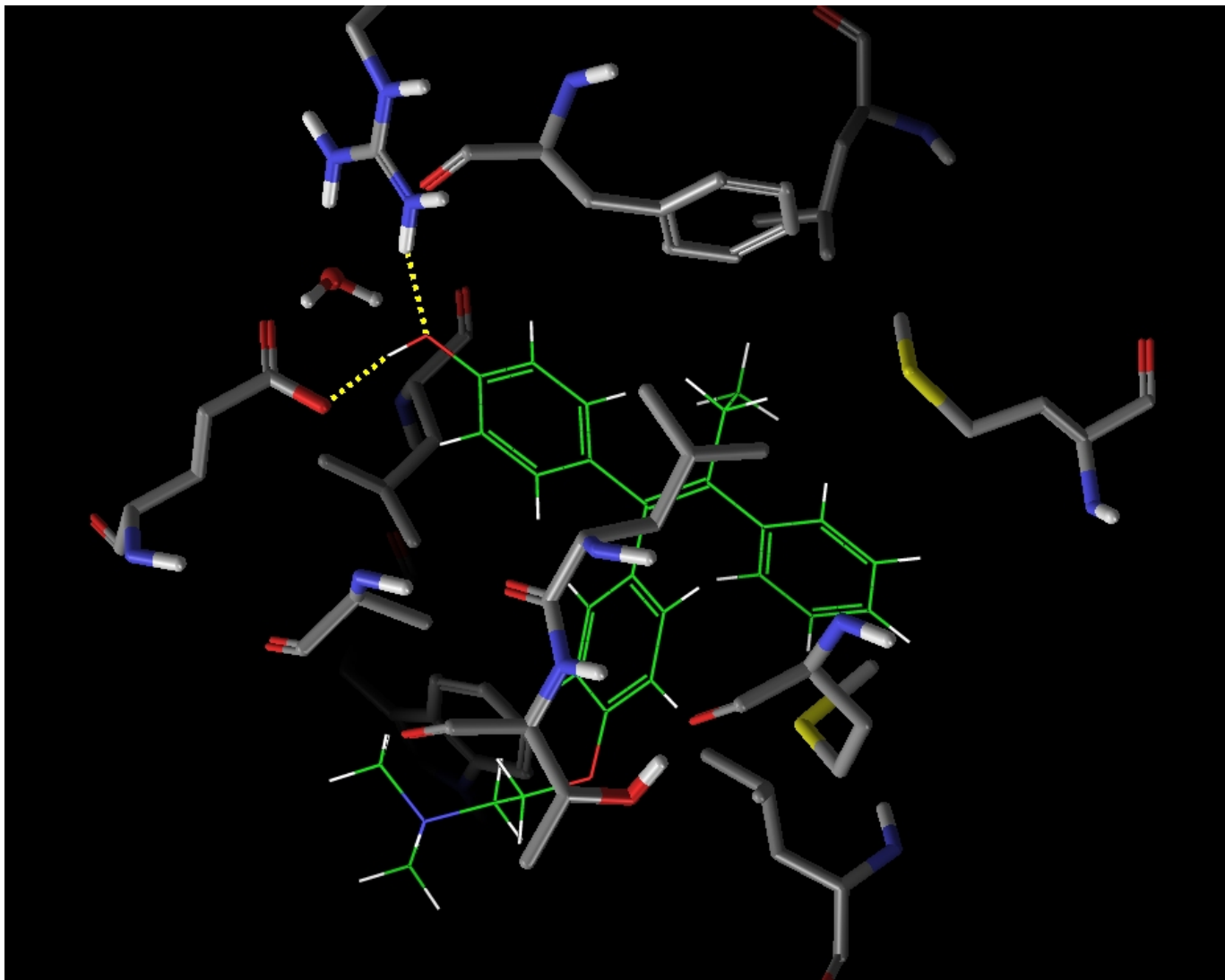
- Use playback controls to step through docking results



- Visualization options can be found under the Entry > View Poses in the Project Table menu bar



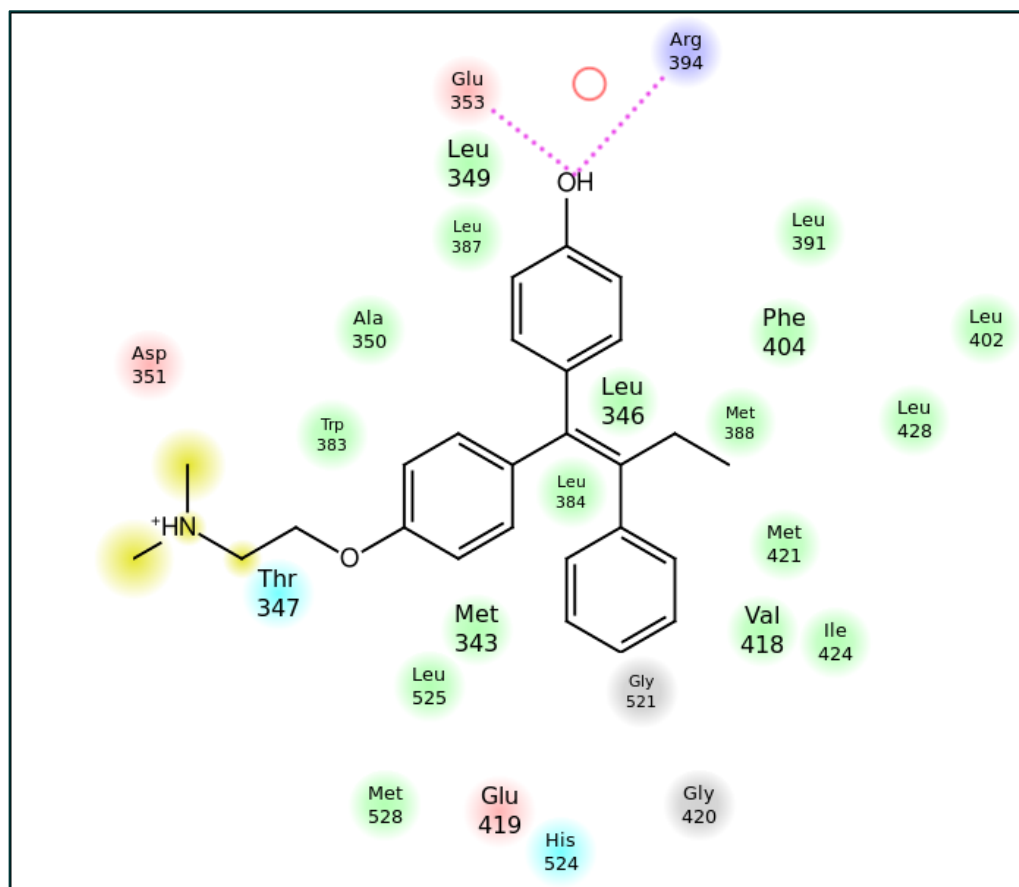
Visualizing H-bonds



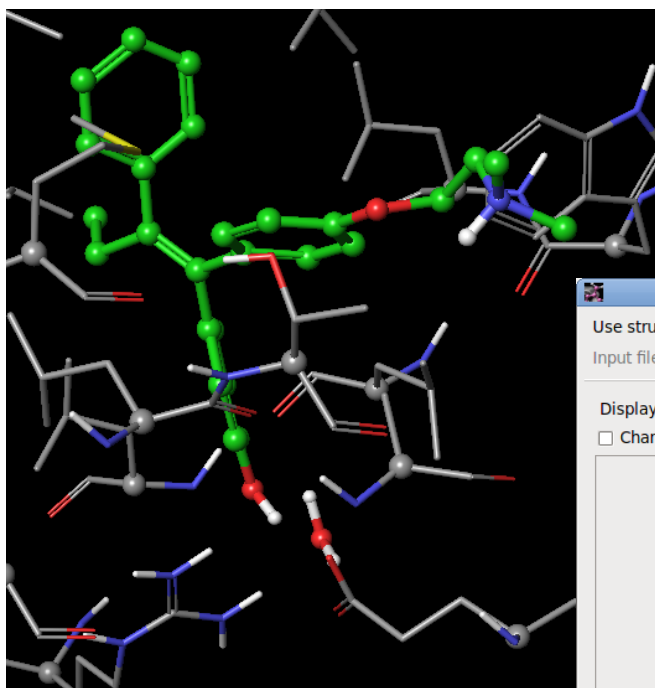
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Creating a Ligand Interaction Diagram

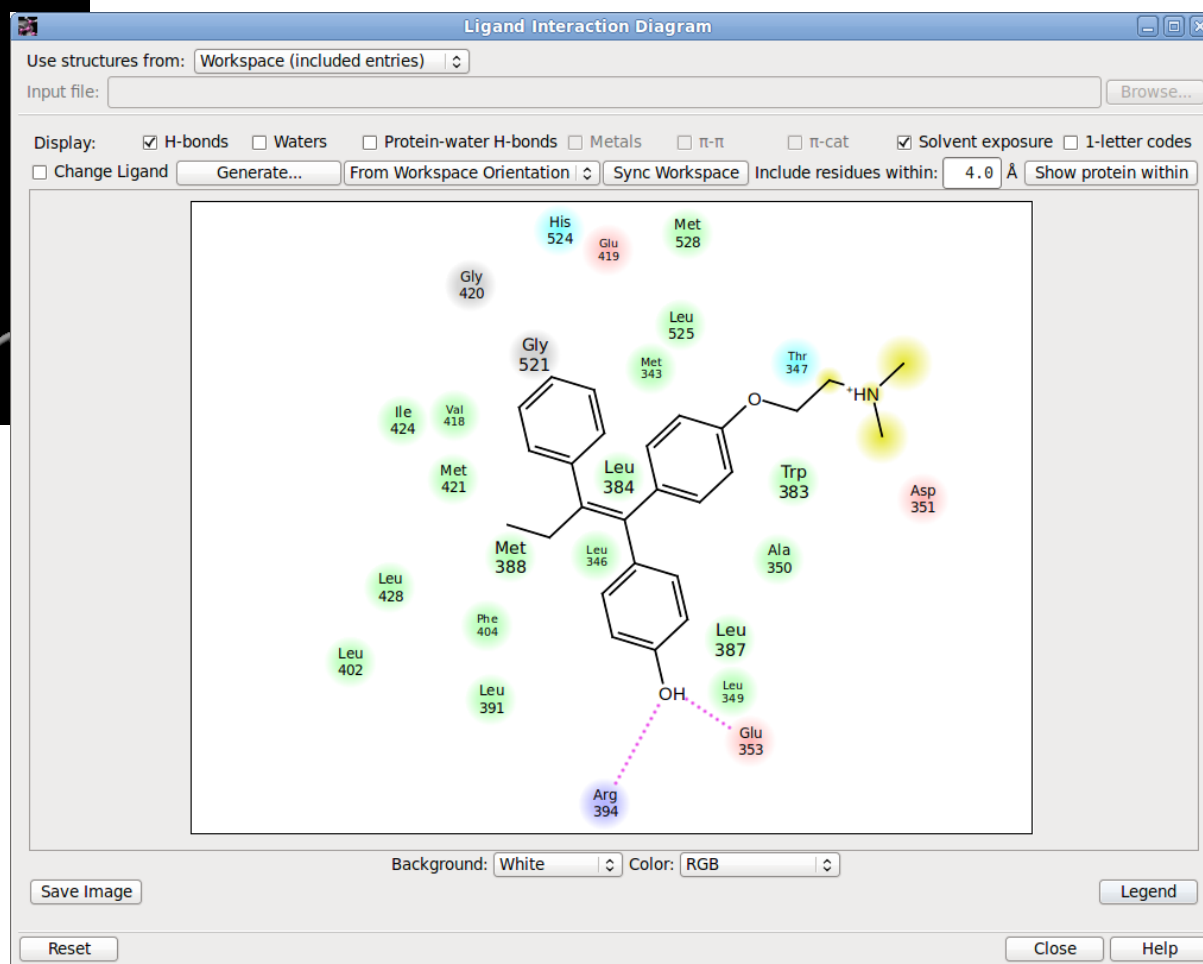
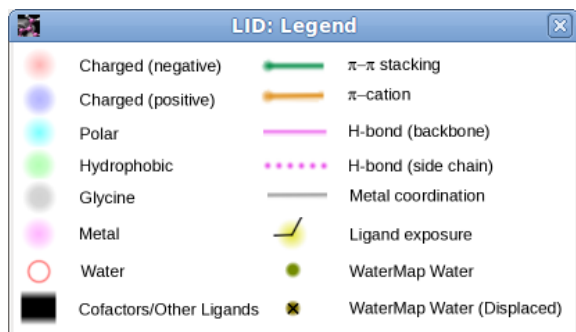
- Creates a 2D representation of ligand binding mode
- More info:
 - <http://www.schrodinger.com/newsletter/17/104/>



Ligand Interaction Diagram for 3ERT

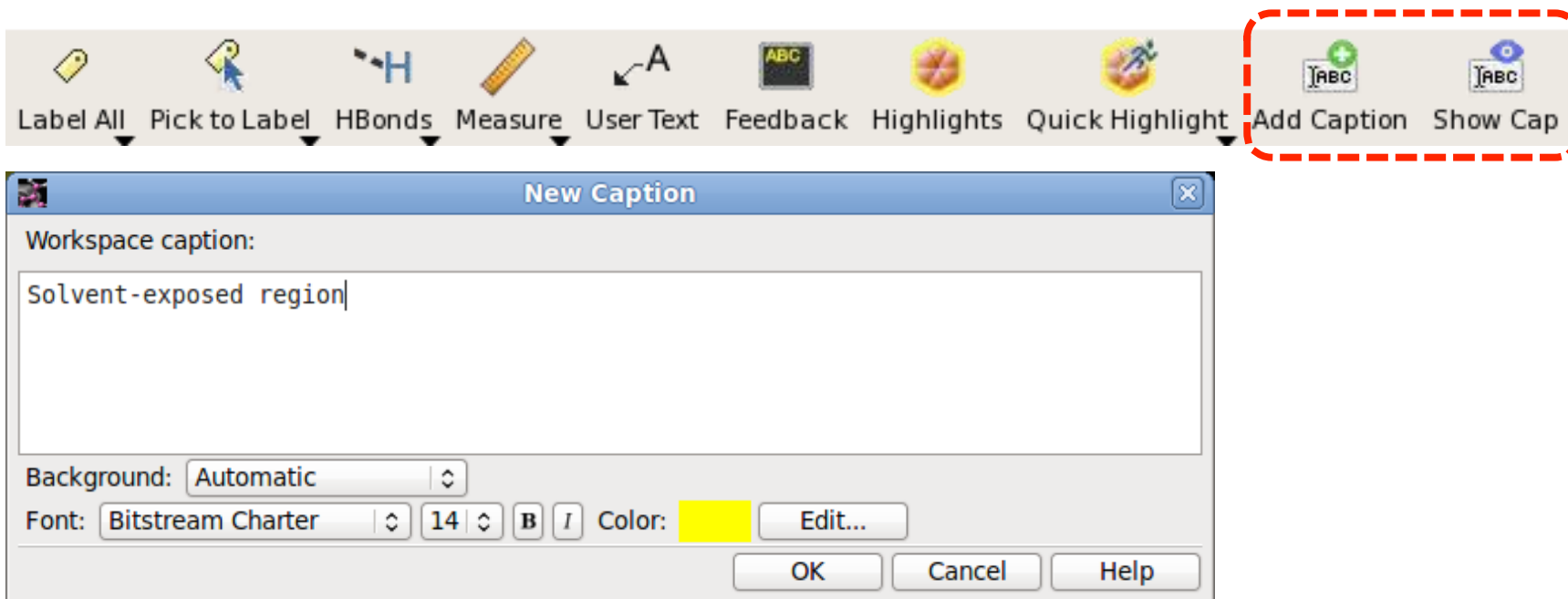


- Orientation of 2D image matches the orientation of 3D ligand in the Workspace.

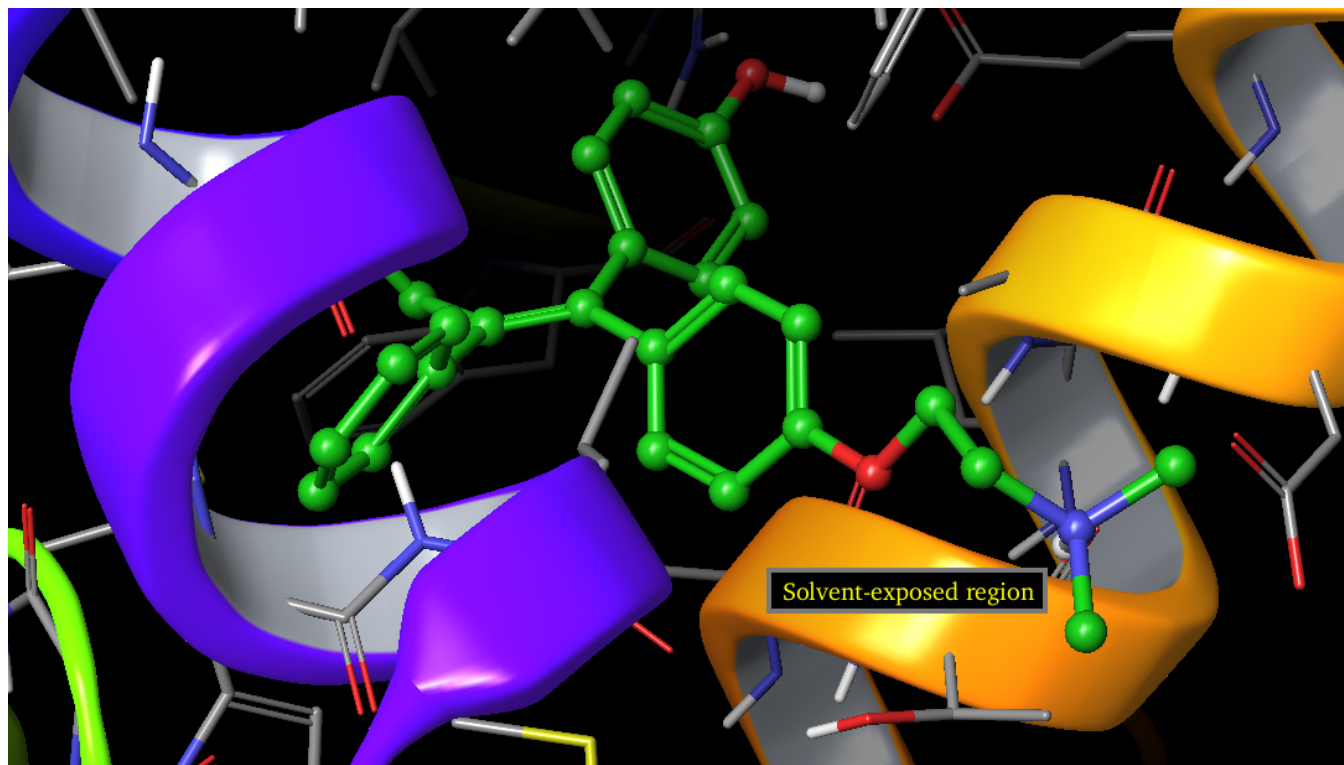


Annotations

- Click “Add Caption” in the Labels Toolbar.
- Click and drag to move annotations in the Workspace.
- Click “Show Captions” in the Labels Toolbar to toggle annotations on and off.



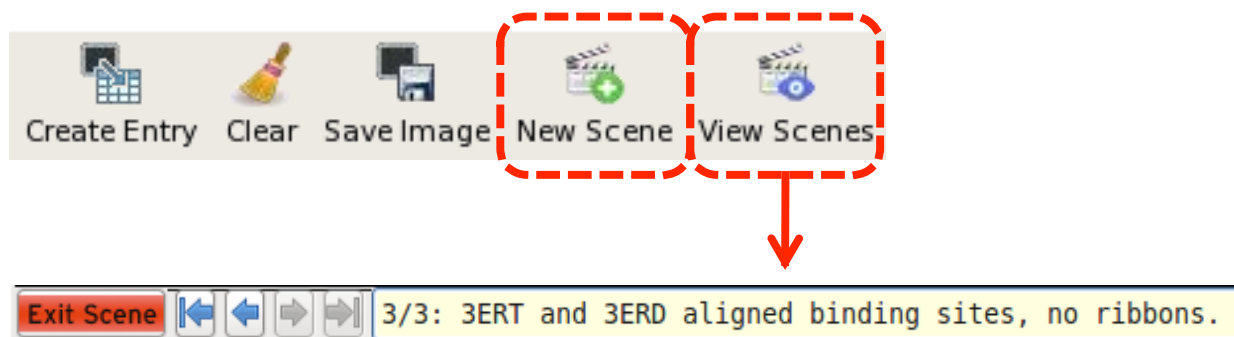
Annotating the Workspace – Example caption



- Click on a caption and press “Delete” to remove it.
- Annotations persist in the Workspace – Unlike atom labels, they are not associated with an individual structure.

Saving AND SHARING scenes in Maestro

- Saving a scene will save all structures and their exact appearance in the Workspace.
- Scenes are useful when you want to save a “snapshot” of the Workspace contents.
- Use the Workspace toolbar to create and view scenes.
- Clicking the “View Scenes” toolbar button lets you step through saved scenes with a special toolbar.



Your new skills

- Familiarity with Maestro user interface and some popular tools
- Knowledge of how to set up and run a self-docking job with Glide to validate a target model
- Familiarity with a handful of docking analysis tools
- Confidence to explore additional tools for virtual screening, addressing receptor flexibility, and other computational tools to address your research questions

Thank you, and good luck!

- Contact Me: tara.phillips@schrodinger.com
- Contact Kristin: kristin.robinson@schrodinger.com
- Newsletters and Seminars are great resources
 - schrodinger.com/seminarcenter
 - schrodinger.com/newsletter
- The Knowledge Base has answers to many common questions
 - schrodinger.com/kb
- Our support team is committed to your success
 - help@schrodinger.com